Settlement Ether and the Next Evolutionary Step in Cryptographically Stored Value

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Abstract

We introduce *Settlement Ether*, a third-generation blockchain architecture that emerges naturally from the technique of *cryptographic ownership separation* introduced in European Patent n°EP26300368.3 (2026) [42, 41], and demonstrated in production by the Lemnicoin (LEMNI) cryptocurrency on Polygon mainnet, and applied to Bitcoin itself through a two-layer dual-authority construction built entirely on long-activated Bitcoin features—SegWit v0 (P2WSH) for quantum-safe custody at rest and Taproot/Tapscript for spending—requiring no change to the Bitcoin protocol. We position this work within an evolutionary trajectory of cryptocurrency: Bitcoin (Generation 1.0) demonstrated that value can be made cryptographic; Ethereum (Generation 2.0) demonstrated that cryptographic value can be made programmable; cryptographic ownership separation enables Generation 3.0, in which the chain itself need not be the source of ownership truth. We argue that this third evolutionary step is being compelled by the quantum computing threat: the conventional response—chain-level migration to post-quantum cryptography—involves coordination costs untenable within the timeline contemporary quantum-hardware projections demand, while cryptographic ownership separation provides post-quantum protection unilaterally, and, in doing so, reveals a deeper architectural inversion of which quantum resistance is only one consequence. We perform a layer-by-layer audit of the quantum vulnerability surface of contemporary blockchains and establish that the threat reduces, in practice, to a single cryptographic primitive: the signature scheme that authorizes value transfer. We then show that this single vulnerability can be remediated by separating the ownership-authorization signature from the network-validity signature, placing only the former on a quantum-resistant cryptographic basis. The consequences for blockchain architecture are profound: the chain becomes a pure settlement and ordering layer; state can be moved off-chain; block-based consensus becomes optional; throughput improves by orders of magnitude. We argue that the migration to Generation 3.0 does not require coordinated protocol changes at the chain level; it can proceed unilaterally, by users adopting the patented, licensable smart contract approach like Lemnicoin, or the dual-authority output construction, on any chain that supports P2WSH for quantum-safe custody at rest and Taproot/Tapscript, or equivalent programmable spending conditions.

Keywords: post-quantum cryptography; blockchain architecture; cryptographic ownership separation; dual-authority signatures; settlement layer; Taproot; Tapscript; Lemnicoin; LAIP; third-generation blockchain; Falcon; Dilithium; SPHINCS+; quantum resistance

1 Introduction

The history of cryptocurrency, regarded as the history of a technical artifact, can be summarized in two transitions. Each redefined what a blockchain is for; each was a categorical, not incremental, change.

The first transition was the publication of the Bitcoin whitepaper in 2008 and the launch of the Bitcoin network in 2009 [1]. Bitcoin demonstrated, for the first time, that value could be made *cryptographic*—that the right combination of public-key cryptography, hash chaining, and proof-of-work could produce a distributed, censorship-resistant store of value, requiring no central authority. Money itself, Bitcoin showed, is a special case of information; and information can be authoritative without being centrally administered. This was Generation 1.0 of cryptocurrency: *cryptographic value*.

The second transition was the launch of Ethereum in 2015 [2, 3]. Ethereum demonstrated that the techniques of cryptographic value could be extended to general computation. A Turing-complete virtual machine on the chain enabled programmable contracts; from these emerged the entire ecosystem of decentralized applications: decentralized finance, non-fungible tokens, decentralized autonomous organizations. This was Generation 2.0: *programmable cryptographic value*.

Both generations share a defining architectural commitment: **the chain is the source of ownership truth.** Whether a UTXO can be spent, whether a smart contract function can be invoked, whether an account has a positive balance—all of these questions are answered by the chain itself, with the chain's native signature scheme acting as the cryptographic substrate of ownership. The chain is the universal authority on who owns what.

This paper argues that this commitment is contingent, not essential, and that relaxing it produces a third evolutionary step in cryptocurrency. We introduce **Settlement Ether**, a third-generation blockchain architecture in which the chain provides ordering and settlement of transactions but does not authoritatively validate ownership. Ownership is instead established by separate cryptographic keys—*ownership keys*—that the chain need not know about and need not validate. The chain's role narrows from "deciding what is true" to "ordering what happened." The architectural consequences are far-reaching: the chain becomes minimal, throughput improves dramatically, and—most consequentially for the present moment—quantum resistance becomes a property of individual users' key choices, rather than a property the chain must coordinate at the protocol level.

This narrowing of the chain's role is not merely an efficiency optimization; it is a precondition for the scale of commerce the coming decades will demand. A pure settlement-and-ordering layer, relieved of the burden of validating ownership on chain, can sustain throughput orders of magnitude beyond what block-based consensus permits—the throughput that legions of autonomous software agents, transacting continuously and without human mediation, will require. The volume and variety of such activity—machine-negotiated contracts, real-time settlement of services rendered between agents, micropayments, and every other form of commercial exchange—will outstrip what any chain that must itself adjudicate ownership can serve. Settlement Ether is built for that regime. And because ownership is established off the chain, *ownership—where it is germane to a transaction at all—is constituted by whatever cryptographic means the transacting parties themselves elect*: a hash-based possession proof, a lattice or isogeny signature, a threshold or multi-party scheme, a primitive not yet standardized, or any composition thereof. The chain neither prescribes nor needs to comprehend the cryptographic basis of ownership; it orders what happened, and the owners decide what owning means. Quantum resistance is then merely the first and most urgent instance of a far more general freedom.

That this third evolution is occurring now—rather than as a gradual architectural drift over decades—is a consequence of the quantum computing threat. The advent of cryptographically relevant quantum computers compels a re-examination of every cryptographic primitive on which blockchain rests. The conventional response, namely the migration of each chain to post-quantum cryptography at the protocol level, involves coordination costs that are, for chains with conservative governance, untenable within the timeline that contemporary quantum-hardware projections demand. The dual-authority construction we describe in this paper emerged from

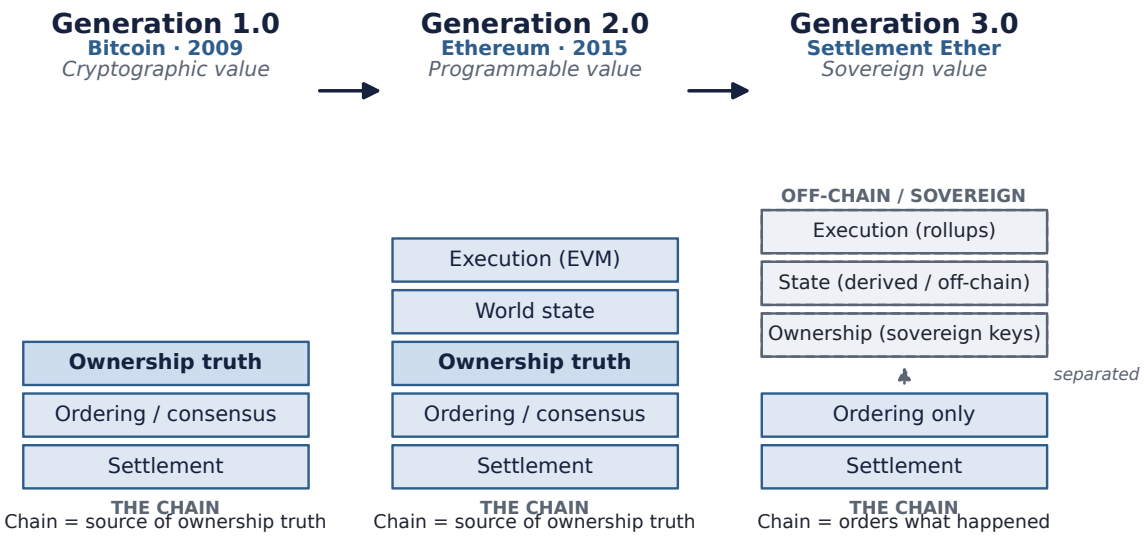


Figure 1: Three generations of cryptocurrency. Each generation redefines the relationship between the chain and the truth it stores. Generation 3.0 emerges from cryptographic ownership separation.

the need to provide post-quantum protection without waiting for protocol-level coordination. Its architectural implications, however, extend far beyond quantum resistance. The pressure that produced the construction was quantum; the architectural inversion the construction enables is more general. Generation 3.0 is, in this sense, a *forced* evolutionary step—an architectural form that the cryptocurrency industry had not previously needed to reach, but that the quantum threat has compelled. Were it not for quantum, the present architecture might have persisted for decades. Because of quantum, it cannot.

The technique that enables this architectural inversion is *cryptographic ownership separation*, the subject of European Patent n°EP26300368.3 (2026) [42], filed by the present author as a continuation-in-part of an earlier filing covering account-based blockchain platforms [41]. We have implemented and demonstrated this technique in the Lemnicoin (LEMNI) cryptocurrency on Polygon mainnet (contract `0x02011c7B7D9C6dfc563c92197557e8BE490a85b4`), which uses a dual-authority model with a quantum-resistant ownership key bound to the on-chain ECDSA wallet key via overridden ERC-20 interface functions. We have further extended the technique to Bitcoin via Tapscript (BIP-341, BIP-342) [9, 10], providing a path by which any Bitcoin user can obtain quantum-resistant fund protection without any change to the Bitcoin protocol itself.

The paper proceeds as follows. Section 2 reviews the architectural commitments of Generations 1.0 and 2.0 and the quantum threat that those architectures face. Section 3 audits the quantum vulnerability surface of current blockchains and establishes that the threat reduces, in practice, to a single cryptographic primitive. Section 4 introduces cryptographic ownership separation as a technique and describes its implementation on account-based chains (LEMNI) and UTXO-based chains (the Bitcoin Tapscript construction). Section 5 derives the implication: that only the ownership key need be quantum-resistant, and that this property is achievable unilaterally without protocol coordination. Section 6 develops the Settlement Ether architecture in detail. Section 7 analyzes throughput improvements. Section 8 positions Settlement Ether within the evolutionary trajectory of cryptocurrency. Section 9 considers migration paths. Section 11 situates the present work against other proposed quantum-resistance and account-abstraction approaches. Section 12 concludes.

2 Background and the Quantum Threat

2.1 The First-Generation Architecture: Bitcoin

Bitcoin's design [1] consists of three interlocking cryptographic mechanisms. First, transactions are signed using ECDSA over the secp256k1 elliptic curve [8]; a transaction is valid if and only if it carries a signature verifiable against the public key whose hash forms the spending address. Second, transactions are aggregated into blocks, each of which contains the hash of the previous block, producing a tamper-evident hash chain. Third, blocks are produced by a proof-of-work consensus mechanism in which miners compete to find a nonce that hashes the block below a target. The combination of these mechanisms yields a distributed ledger in which historical entries cannot be feasibly altered and in which spending requires possession of a specific cryptographic private key.

Bitcoin's role in the evolution of cryptocurrency is foundational. It demonstrated that value could be made cryptographic, that a distributed ledger could be maintained without any central authority, and that the security of the system could rest entirely on the difficulty of cryptographic problems. The UTXO model that Bitcoin introduced—in which transaction outputs are first-class objects that can be spent at most once—remains a foundational paradigm in chain design.

2.2 The Second-Generation Architecture: Ethereum

Ethereum [2, 3] extends Bitcoin's architecture in one critical direction: the chain stores not only balances but also *state*, and not only state but *code that operates on state*. The Ethereum Virtual Machine (EVM) is a Turing-complete execution environment in which programs—*smart contracts*—can be deployed and called by users. The chain maintains a *world state*, a global mapping from addresses to balances and contract storage, updated atomically by every transaction. The signature scheme is essentially the same as Bitcoin's (ECDSA over secp256k1), but the architectural commitment has been considerably extended: the chain is now responsible not only for recording authoritative ownership of fungible value but also for executing arbitrary programs that mediate the transfer of that value.

The architectural commitment that the chain is the source of ownership truth becomes, in the Ethereum context, a much heavier one. The world state grows monotonically; the EVM must be executed identically on every full node; the gas accounting system must price all operations consistently; and any change to the protocol must propagate through a population of independent client implementations.

2.3 Account Abstraction and the Rollup-Centric Roadmap

Both Bitcoin and Ethereum have, in their subsequent evolution, taken steps that complicate the strict "chain as source of truth" commitment. Ethereum's account abstraction proposals (ERC-4337 [13], EIP-7701 [14]) allow smart-contract-wallet accounts whose verification logic is arbitrary code rather than a fixed signature check. The rollup-centric roadmap [16] pushes bulk transaction execution to higher layers, with the base chain functioning primarily as a settlement and data-availability layer for compressed proofs from those higher layers. Bitcoin's Taproot upgrade [9, 10] introduced script trees that enable complex spending conditions while preserving on-chain privacy for the common case.

These evolutionary steps are best understood as partial relaxations of the architectural commitment: the chain is still the source of ownership truth, but the means by which ownership is asserted have become more flexible. The present work argues that this trend, taken to its logical conclusion, produces an architecture in which the chain ceases to be the source of ownership

truth at all.

2.4 The Quantum Threat to the Current Architectures

The cryptographic primitives that Generations 1.0 and 2.0 rely on are vulnerable to attack by a sufficiently capable quantum computer. Shor's algorithm [4], executed on a fault-tolerant quantum processor of sufficient scale, solves the discrete logarithm problem in polynomial time. This breaks ECDSA, Schnorr signatures, EdDSA, and BLS signatures—in other words, the signature scheme of every major blockchain in current use. The hardware required to execute Shor's algorithm against a 256-bit elliptic curve key has been estimated by Google Quantum AI in March 2026 at fewer than 500,000 physical qubits with surface-code error correction [28], a target that contemporary roadmaps from IBM, Quantinuum, and IonQ project to be reached by the late 2020s or early 2030s [29, 30, 31].

The threat is not merely future. As established by Mascelli and Rodden in a September 2025 Federal Reserve Board paper [25], the public nature of blockchain ledgers makes them uniquely susceptible to *harvest-now-decrypt-later* (HNDL) attacks. Every transaction signature ever broadcast on a blockchain is, by design, public; every public key ever revealed by a spent UTXO or transacted account is, by design, public; an adversary who in the future obtains a cryptographically relevant quantum computer (CRQC) can retrospectively recover the corresponding private keys and use them to forge transactions that the chain will accept as cryptographically valid. The harvest is operational today even though the decryption capability does not yet exist.

The blockchain industry has responded with a range of proposed migration mechanisms. NIST's post-quantum cryptography standardization process [17] has produced three primary standards (ML-KEM/FIPS-203 [18], ML-DSA/FIPS-204 [19], SLH-DSA/FIPS-205 [20]) with a fourth (FN-DSA/Falcon [21]) in development and a fifth backup KEM (HQC [22]) selected in March 2025. Bitcoin proposals BIP-360 and BIP-361 [11, 12] would activate post-quantum signature schemes as new spending categories. Ethereum's roadmap incorporates EIP-7701 and EIP-8141 for native account abstraction and post-quantum transaction formats. Solana's April 2026 announcement [32] committed to a Falcon-based migration. StarkWare's Quantum Safe Bitcoin (QSB) proposal [26] uses STARK proofs of hash preimage knowledge as a wrapper construction.

All of these proposals share a feature: they require protocol-level coordination. A chain that adopts any of them must execute a hard fork or activation event, with all the political and technical friction that implies—which, on chains with conservative governance such as Bitcoin, may take years or may not happen at all. The present work proposes a different path: one in which quantum-resistant fund protection is achievable unilaterally by individual users, without any change to the chain's protocol, on any chain that supports programmable spending conditions of the Tapscript family.

3 The Quantum Vulnerability Surface of a Blockchain

To make precise the claim that the quantum threat to blockchain can be remediated at a single point, we audit the cryptographic primitives of a contemporary blockchain layer by layer. We ask, for each layer: what quantum algorithm threatens this primitive, and what is the consequence of the threat?

3.1 The Two Quantum Algorithms That Matter

Of the broad space of quantum algorithms studied over the past three decades, only two pose meaningful threats to blockchain cryptography. Shor's algorithm [4] reduces the discrete logarithm problem (DLP) and the integer factorization problem (IFP) to problems solvable in polynomial time on a quantum computer. This breaks every public-key signature scheme whose security rests on DLP or IFP, including RSA, the Diffie–Hellman key exchange and its elliptic-curve variants, ECDSA, Schnorr signatures, EdDSA, and the BLS pairing-based signatures used in Ethereum's validator-attestation aggregation. The break is total: given a public key, a quantum computer of sufficient scale recovers the corresponding private key in time polynomial in the key length.

Grover's algorithm [5] provides a quadratic speedup on unstructured search. Applied to a cryptographic hash function, it reduces the effective security of a b-bit hash from 2^b classical operations to approximately $2^{b/2}$ quantum operations. For SHA-256, the bit-strength is reduced from 256 to approximately 128, which remains well beyond computationally feasible.

No other quantum algorithm currently known materially weakens any other component of a blockchain. Quantum simulation, quantum optimization, and quantum machine learning—while potentially significant for other applications—do not threaten blockchain cryptography. The quantum threat reduces, in scope, to Shor against the signature schemes and Grover against the hash functions.

3.2 Layer-by-Layer Audit

Figure 2 summarizes the quantum exposure of the cryptographic layers of a contemporary blockchain. We discuss each layer in turn.

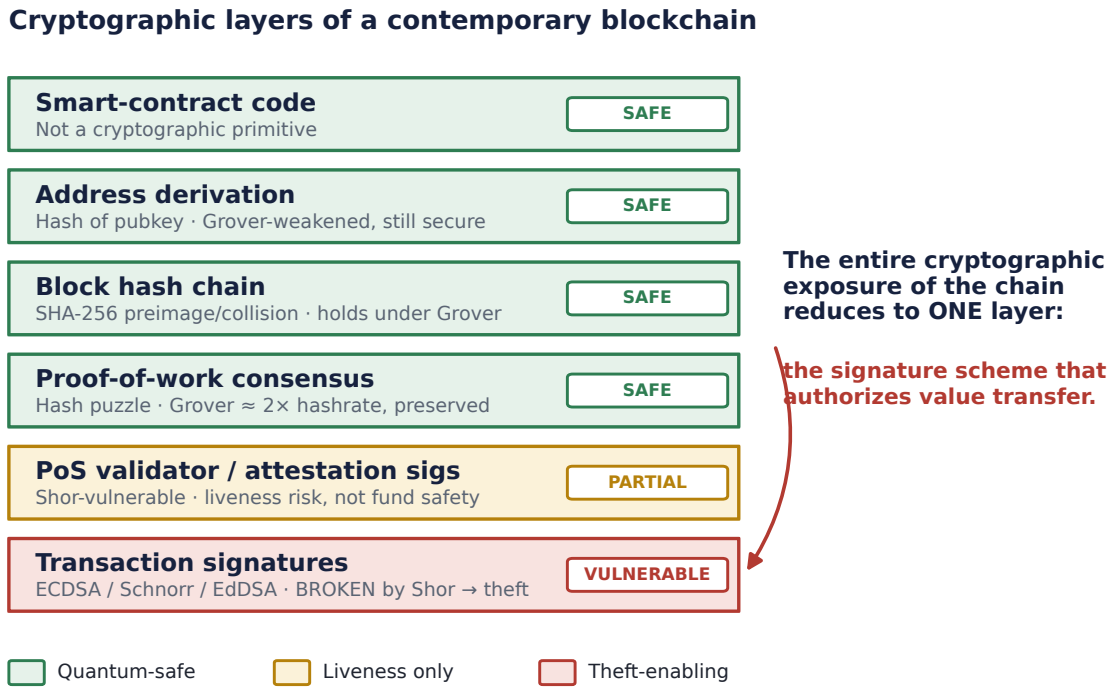


Figure 2: The quantum vulnerability surface of a contemporary blockchain. The cryptographic exposure of an entire blockchain reduces, in practice, to its transaction signature scheme.

Smart contract code is not cryptographic. Once deployed, a smart contract is simply a sequence of bytecode instructions executed by every full node. The quantum threat does not apply.

Address derivation, on chains that hash public keys to produce addresses (Bitcoin P2PKH, Ethereum's Keccak truncation), is a hash operation. Grover halves the effective bit-strength, but the resulting margin remains computationally secure for any feasible quantum capability. Addresses themselves are quantum-safe; the vulnerability lies in what they derive from, which is examined separately below.

The block hash chain—the chain of cryptographic hashes that links each block to its predecessor—rests on the preimage and collision resistance of the underlying hash function (typically SHA-256). Grover weakens both, but the chain remains computationally secure. The integrity of the historical record is, broadly, preserved.

Proof-of-work consensus (Bitcoin) is a hash-based puzzle. Grover halves the effective work factor required to find a valid nonce, which is equivalent to a doubling of mining hashrate. The chain continues to function; the competitive dynamics among miners shift, but the consensus property is preserved.

Proof-of-stake validator and attestation signatures (Ethereum since 2022, Solana, Cosmos, Polkadot, and others) rest on Shor-vulnerable signature schemes. A quantum adversary that compromises validator keys can forge attestations, attempt double-spends, or disrupt consensus liveness. This is a real concern, but it affects *liveness* rather than *user fund safety*: individual users' funds are protected by their transaction signatures, not by validator attestations. A compromised validator set is a serious operational problem, but it is, in principle, separable from the user-fund-safety problem.

Transaction signatures (ECDSA, Schnorr, EdDSA, depending on the chain) are completely broken by Shor's algorithm. This is the single point at which the quantum threat directly enables theft: an adversary with a CRQC, presented with the public key associated with an account, can recover the private key and use it to sign transactions that the chain accepts as valid.

3.3 The Narrowness of the Threat

The cumulative result of the audit is striking. Of the cryptographic mechanisms that compose a blockchain, only one—the signature scheme that authorizes value transfer—is meaningfully broken by a CRQC in a way that enables direct theft of funds. The hash chain holds. The address derivation holds. The proof-of-work continues. The smart contract code is not a cryptographic primitive in the first place. Even the proof-of-stake validator layer, while quantum-vulnerable, presents a liveness rather than a fund-safety concern.

This narrowness is the central technical observation of the present paper. The conventional framing—that "blockchains must migrate to post-quantum cryptography"—is, on closer inspection, overbroad. The precise statement is: *the signature scheme that authorizes value transfer must be migrated to post-quantum cryptography.* Everything else in a blockchain remains quantum-safe under any plausible projection of quantum hardware development.

If the migration could be confined to that single point—if the signature scheme that authorizes value transfer could be replaced without replacing the chain's native signature scheme—then the entire coordination problem of protocol-level post-quantum migration would dissolve. The chain could remain exactly as it is. Only the authorization mechanism would change. And that change could, in principle, be made by individual users, unilaterally, without requiring any consensus from miners, validators, or developers.

This is precisely what cryptographic ownership separation accomplishes.

4 Cryptographic Ownership Separation

4.1 The Dual-Authority Model

Cryptographic ownership separation [41, 42] replaces the single signature that authorizes value transfer with two structurally distinct signatures drawn from different cryptographic schemes. Specifically:

- The **wallet key** is a standard signature scheme native to the underlying chain (e.g., ECDSA over secp256k1 on Bitcoin and EVM-based chains, Ed25519 on Solana). It satisfies the chain's transaction-validity requirements at the network protocol level.
- The **ownership key** is a separate signature, drawn from an independent cryptographic scheme that may be—and in the present construction *is*—quantum-resistant. Candidate schemes include Falcon (FN-DSA), Dilithium (ML-DSA), SPHINCS+ (SLH-DSA), or hash-based one-time-signature schemes such as Lamport or Winternitz.

Both signatures must be present for a transaction to be authoritative. Neither alone is sufficient. The chain's protocol layer enforces the wallet-key signature in the standard way; the ownership-key signature is enforced by an additional verification mechanism layered on top of the chain's existing capabilities. The two cryptographic schemes operate at different layers and serve different functions: the wallet key satisfies the network; the ownership key proves actual ownership.

4.2 Implementation on Account-Based Chains: Lemnicoin

On account-based chains supporting smart contracts (Ethereum, Polygon, Solana, Cosmos, Polkadot, NEAR, Aptos, Sui, and others), the dual-authority model is implemented by overriding the standard token-interface functions to require ownership-key verification in addition to the chain's native signature check [41]. For an ERC-20 token, for example, the `transfer` and `transferFrom` functions are overridden such that they require a valid ownership-key signature over the transaction's parameters, verified on-chain against the ownership-key commitment that was irrevocably bound to the account at registration time.

The Lemnicoin (LEMNI) cryptocurrency demonstrates this construction in production on Polygon mainnet (contract `0x02011c7B7D9C6dfc563c92197557e8BE490a85b4`). LEMNI uses Falcon-512 as the on-chain quantum-resistant ownership signature scheme, verified by the ZKNOX ETHFALCON Solidity library. The chain's native wallet key is a standard secp256k1 ECDSA key; spending requires both signatures. The architecture uses the Diamond proxy pattern (EIP-2535) [15] to enable cryptographic agility—the verification logic can be upgraded to support new post-quantum schemes (or to replace existing schemes that fail cryptanalysis) without disturbing token holders, who retain their funds and their ownership keys across upgrades.

The Lemnicoin implementation incorporates an additional cryptographic layer, the Lemniscate-AGM Isogeny Protocol (LAIP), which provides a deeper isogeny-based identity binding bound to the Dilithium5/ML-DSA-87 standardized lattice signature [40]. A full treatment of LAIP is beyond the scope of the present paper, which addresses cryptographic ownership separation as an architectural pattern rather than the choice of specific post-quantum primitive. The architectural pattern of the present paper is independent of which post-quantum scheme is selected as the ownership signature: any of Falcon, Dilithium, SPHINCS+, LAIP, or future schemes will serve equally well.

4.3 Implementation on UTXO-Based Chains: Bitcoin via Tapscript

For UTXO-based chains without general smart contract execution, the dual-authority model requires a fundamentally different implementation strategy [42]. The extension to Bitcoin uses the Taproot output format (BIP-341) [9] and Tapscript leaf scripting (BIP-342) [10] to implement the ownership-key check as a native spending condition.

The construction uses two layers, both derived from a single seed. At rest, funds are held in a SegWit version 0 pay-to-witness-script-hash (P2WSH) output. The on-chain commitment of a P2WSH output is only the SHA-256 hash of its witness script: no public key and no key-path spending condition appear on chain. The witness script enforces the dual-authority condition—a wallet-key signature check (`OP_CHECKSIGVERIFY` against the wallet's compressed public key) followed by a hash-based ownership *possession proof*, a sequence of `OP_SHA256`/`OP_EQUALVERIFY` checks against committed hashes—with transaction binding supplied by the co-required wallet signature and enforced one-time use of each protected address. To move funds, the holder migrates the P2WSH output into a SegWit version 1 (Taproot) output of the established dual-authority form—internal key set to a Nothing-Up-My-Sleeve (NUMS) point, the dual-authority condition carried in a Tapscript leaf—and spends from there. Wallet and ownership material for both layers derive deterministically from the same seed and address index, so the two layers are two views of one protected position rather than two independent custody arrangements. Bitcoin's legacy per-script opcode limit caps the rest-layer possession proof at roughly one hundred committed hashes; the Tapscript spend layer, which BIP-342 frees of that limit, carries the full-length proof. Each committed hash is an independent preimage of cryptographic width, so the proof's strength does not depend on carrying a large number of them.

The two-layer structure is forced by a subtlety a Taproot-only construction does not survive. A Taproot output commits, on chain and in the clear, to a single curve point: the tweaked output key Q. Setting the internal key to a NUMS point makes key-path spending impossible *classically*, because no entity knows the discrete logarithm of Q; it does *not* make key-path spending impossible against a CRQC, because Shor's algorithm recovers the discrete logarithm of Q directly from the on-chain point, irrespective of how Q was constructed. An adversary who recovers it spends via the key path, bypassing the dual-authority script entirely. A Taproot dual-authority output therefore protects against classical key theft but leaves an *idle* output's key exposed to a quantum harvester—precisely the harvest-now-decrypt-later exposure of Section 3. The P2WSH rest layer removes this exposure at its source: with only a script hash on chain and no key-path condition, a quantum harvester observing an unspent output has no public key to attack. The wallet public key and the ownership commitments enter the chain only in the witness, at the moment of the migration spend.

The ownership factor is verified within Tapscript using either (i) a hash-based *possession proof* in the current leaf version, in which the owner reveals preimages of committed hashes and the script checks them with only `OP_SHA256` and `OP_EQUALVERIFY`—transaction binding being supplied by the co-required Schnorr signature, since stock Tapscript cannot bind a hash-based proof to the transaction message; (ii) a message-bound hash-based signature such as Lamport, Winternitz (W-OTS+), or SPHINCS+ (SLH-DSA), once `OP_CHECKSIGFROMSTACK` or `OP_CAT` is activated; or (iii) a future leaf version that activates native opcodes for Falcon, Dilithium, or generic post-quantum signature verification. Critically, the possession-proof construction under method (i) is available *today*, requiring no soft fork, hard fork, or protocol change: P2WSH (SegWit v0, 2017) and Taproot (2021) are both long-activated, and the rest-layer and spend-layer transactions are standard outputs that any Bitcoin node validates normally. One residual remains. The migration spend, like every Bitcoin spend, transiently reveals the wallet public key in the confirmation window, where a CRQC could in principle recover the wallet key and front-run the spend before it confirms. This window is shared by every Bitcoin transaction, not specific to the dual-authority construction; it is mitigated operationally by submitting the spend out-of-

band to an identifiable miner and cryptographically by combination with a transaction-integrity scheme such as QSB, and is eliminated natively once `OP_CAT` or `OP_CHECKSIGFROMSTACK` permits a message-bound ownership proof (Section 11). Funds resting in the P2WSH layer are never subject to this window; only the act of moving them is.

4.4 The Patent

The cryptographic ownership separation technique, the dual-authority output construction, the NUMS-internal-key construction that closes the key-path bypass, and the migration path to native post-quantum opcodes are all subjects of European Patent n°EP26300368.3 (2026) [42], filed by the present author as a continuation-in-part of an earlier filing covering the account-based implementation [41]. The present paper develops the architectural implications of the technique; the patent specifies the technique itself.

5 The Quantum-Resistance Implication

The quantum-resistance implication of cryptographic ownership separation is the corollary of two facts established above. First (Section 3), the quantum threat to a blockchain reduces, in practice, to the signature scheme that authorizes value transfer. Second (Section 4), cryptographic ownership separation replaces that signature with a pair of signatures, only one of which (the ownership key) need be quantum-resistant.

The conclusion is direct: with a dual-authority output, possession of the wallet key alone—whether stolen classically or recovered by a CRQC—never suffices to move funds, regardless of whether the chain itself has migrated to post-quantum cryptography. On account-based chains, where the ownership signature is verified on chain and bound to the transaction, this protection is complete. On Bitcoin, where stock script cannot yet bind the ownership proof to the transaction, the protection is complete at rest and against classical theft, with one shared residual—the owner's own broadcast window—examined in Sections 4 and 11.

5.1 Only the Ownership Key Need Be Quantum-Resistant

Consider an attacker equipped with a CRQC. The attacker presents a dual-authority output's public information—the wallet public key, the script structure, the ownership-key commitment—and asks: what is the maximum advantage the quantum computer provides?

The wallet public key is, by hypothesis, recoverable by the attacker via Shor's algorithm. The attacker therefore obtains the wallet private key. But the wallet private key alone is insufficient: every spend requires the ownership-key signature, which the attacker cannot produce without the ownership private key. The ownership key is, by construction, drawn from a post-quantum signature scheme; no quantum algorithm currently known recovers a Falcon, Dilithium, or SPHINCS+ private key from the public key in polynomial time. The ownership key is, accordingly, quantum-secure.

The result is that the attacker, despite possessing the wallet private key and a fully functional CRQC, cannot spend a protected output at rest. On account-based chains the ownership signature is verified on chain and bound to the transaction's parameters, so the attacker cannot spend at all: the chain's quantum vulnerability is rendered operationally irrelevant, and fund safety is preserved by the ownership key alone. On Bitcoin the same holds for any output sitting unspent—the rest layer places no public key on chain for Shor to attack—and against any classical thief, who cannot recover the wallet key in the first place. The one case the Bitcoin construction does not eliminate is the owner's own spend: migrating or spending a protected

output transiently reveals the wallet key in the confirmation window, a window every Bitcoin spend shares and which the construction mitigates rather than removes (Sections 4, 11). This property—unilateral quantum-resistant protection of funds achievable without protocol-level chain migration—is achievable in practice only through the construction described in Section 4, which is the protected subject matter of European Patent n°EP26300368.3 (2026) [42] and the prior US application [41].

5.2 The Chain Need Not Migrate

This has a profound consequence for the global post-quantum migration problem. The conventional approach assumes that every chain must individually migrate to post-quantum cryptography—that Bitcoin must soft-fork BIP-360 or BIP-361, that Ethereum must coordinate EIP-7701 and EIP-8141, that Solana must complete its Falcon migration, that Cosmos must update Tendermint, that Polkadot must rework Sr25519. Each of these is a multi-year coordination problem in a community with conservative governance.

Under the dual-authority model, the chain's migration becomes optional. A chain that has migrated to post-quantum cryptography is, for users of dual-authority protection, no safer than a chain that has not. The security guarantee is supplied by the ownership key, which the chain need neither know about nor validate. The chain can remain exactly as it is; the user's funds are protected regardless.

A nuance worth noting: the chain's migration is not entirely irrelevant. As discussed in Section 3, proof-of-stake consensus signatures, if compromised, present a liveness concern at the chain level. A chain whose validator layer is quantum-compromised may produce inconsistent attestations, forks, or temporary unavailability. The fund safety of dual-authority-protected users is preserved in such scenarios, but the chain's overall reliability is impaired. The validator-signature migration is therefore relevant for chain-level liveness even when it is not relevant for user-level fund safety. These are separable concerns, and only the former requires protocol coordination.

5.3 Unilateral Protection

The most consequential property of cryptographic ownership separation is that it can be adopted *unilaterally*. A user who wishes to protect their Bitcoin against quantum capability can construct a dual-authority output today, using only long-activated script machinery: a P2WSH rest output (SegWit v0) whose on-chain footprint is a single script hash, migrating to a Taproot dual-authority output (SegWit v1) to spend. The construction requires no soft fork, no protocol change, no consensus from miners, no developer agreement. The user's wallet software constructs the outputs; the user broadcasts them as standard transactions; the Bitcoin network validates them as standard transactions. At-rest protection is active from the moment of confirmation; the only quantum-exposed interval is the brief, user-initiated migration spend.

The Bitcoin construction just described—the two-layer P2WSH-at-rest, Taproot-at-spend dual-authority output, the NUMS-internal-key mechanism in the spend layer, the script designs that enforce the dual-authority condition in each layer, and the per-address-index derivation with enforced one-time use of each protected address that prevents reuse of a revealed ownership secret—is the subject of European Patent n°EP26300368.3 (2026) [42]. Its application to Bitcoin, and to any other UTXO-based chain that adopts Taproot or an equivalent programmable spending model, is protected under that filing. The analogous construction on account-based chains—the smart-contract-level enforcement of the dual-authority condition demonstrated by Lemnicoin and applicable to every smart-contract chain in current production—is protected by the prior US application [41]. The constructions described in this section, on Bitcoin or on

any other chain, are not freely implementable: they are the intellectual property of the present author and their use requires licensing under the cited filings.

The same logic applies, with appropriate adaptation, to every chain that supports either smart contracts (EVM, Solana, Cosmos, Polkadot, NEAR, Move-based chains) or programmable spending conditions of the Tapscript family (Bitcoin and Bitcoin derivatives). On all such chains, individual users can adopt quantum-resistant fund protection without waiting for—and without depending on—the chain's protocol-level migration.

This is the conceptual inversion at the heart of this paper. The post-quantum migration of blockchains has been treated, throughout the policy and technical literature, as a coordination problem [25, 43, 44, 45, 46]. Under the dual-authority model, it ceases to be a coordination problem at all. It becomes a unilateral decision by individual users.

6 Settlement Ether: The Third-Generation Architecture

If quantum resistance is, properly speaking, a property of the ownership-authorization signature rather than a property of the chain—if the chain need not validate ownership at all, because separate ownership keys do that work—then the question naturally arises: what should the chain do? With ownership validation lifted off, what remains?

We argue that the natural answer produces a third-generation blockchain architecture, which we call *Settlement Ether*, that is radically minimal in comparison with the Generation 1.0 and 2.0 architectures.

6.1 The Architectural Inversion

The defining commitment of the first two generations is that the chain is the source of ownership truth. Under cryptographic ownership separation, this commitment is relaxed. Ownership is asserted by the ownership-key holder; the chain merely records the consequences of valid ownership assertions. The chain becomes, in effect, a public ledger that takes orders rather than a system that owns the state.

Figure 3 summarizes the architectural difference between a contemporary Generation 1.0/2.0 chain and a fully evolved Settlement Ether chain.

6.2 What the Chain Still Needs to Do

Strip away everything inessential and three jobs remain.

Ordering. The chain must establish a total order on transactions, so that double-spends can be detected. If transaction A and transaction B both attempt to spend the same UTXO (or update the same account state), the chain must determine which came first. This is the irreducible role of the chain in any distributed value-transfer system.

Availability. The chain must make the ordered transaction log available to all participants. A transaction that has been ordered but is not visible to other nodes is, for practical purposes, not ordered.

Settlement finality. Once the chain commits to an ordering of transactions up to some point, that ordering must not be revisable. This is what makes the chain a settlement layer rather than a temporary ordering service.

Notice what is *not* on this list. The chain need not validate ownership: the ownership keys do that. The chain need not maintain account balances: balances can be derived from the ordered log by anyone who cares to compute them. The chain need not execute smart contracts:

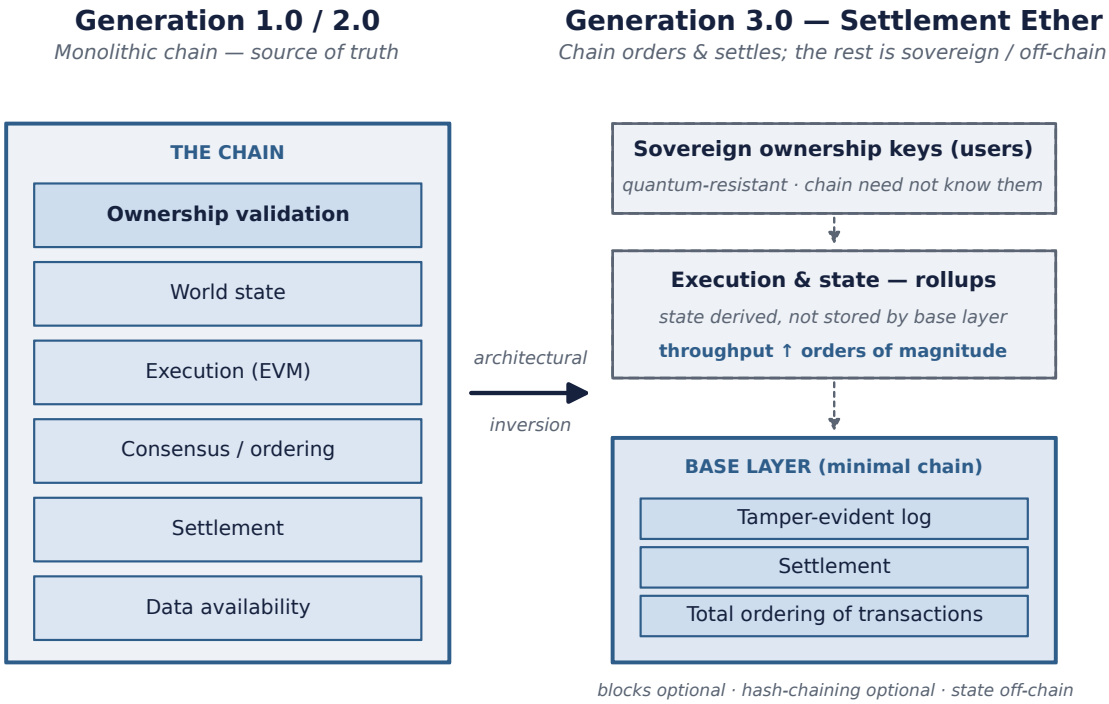


Figure 3: Comparison of contemporary blockchain architecture (Generation 1.0/2.0) with Settlement Ether (Generation 3.0). In Generation 3.0, the chain provides ordering and settlement; ownership, state, and execution are off-chain or sovereign.

contracts can run on higher layers that commit their final states back to the chain. The chain need not even verify the ownership-key signatures: the signatures are useful at the moment of transaction interpretation (when a balance is derived from the log) but not at the moment of ordering (where only the existence of the transaction needs to be recorded).

6.3 Block Structure Becomes Optional

In contemporary chains, transactions are grouped into blocks for a specific reason: proof-of-work and proof-of-stake operate on discrete units. The block is the atomic unit of consensus, and the proof-of-work or attestation aggregation makes a discrete commitment to each block.

In Settlement Ether, the consensus problem is much smaller. The chain is no longer asking "did this person own this asset?"—the ownership keys answered that. The chain is asking "in what order did these transactions arrive?"—which is a much weaker requirement. Several architectural alternatives become available.

Stream-based ordering. Transactions arrive at the chain in a continuous stream, time-ordered and signed by ordering nodes, with no discrete block structure. The chain emits a verifiable sequence rather than a sequence of blocks. Solana's proof-of-history [33] approximates this paradigm.

DAG-based structures. Transactions form a directed acyclic graph, with each new transaction referencing one or more parent transactions. A consensus rule determines which DAG branches become canonical. Hedera [36], IOTA Tangle [37], and Sui's checkpoint mechanism [35] use variants of this.

Pure log-replicated state machines. The chain is an append-only log, replicated across nodes by

a distributed consensus protocol (Raft, Paxos, or a Byzantine fault-tolerant variant), without any block structure at all. Finality is achieved when a quorum of nodes acknowledges each entry.

In all three alternatives, the chain no longer aggregates transactions into discrete units for the purpose of proof-of-work or proof-of-stake. The role of the chain has been reduced sufficiently that the block structure becomes an implementation choice rather than an architectural necessity.

6.4 Hash Chaining Becomes Optional

Closely related to the block question is the question of hash chaining. In a contemporary chain, each block contains the hash of the previous block, producing the tamper-evident structure that makes historical alteration computationally infeasible. The Settlement Ether chain still needs tamper-evidence of the ordering record—an adversary who could rewrite the history of which transaction came first could defeat double-spend protection—but block-header chaining is not the only mechanism that provides it.

Several alternatives exist. A Merkle Mountain Range [38] is an append-only accumulator that produces a single root commitment over an arbitrarily long log; new entries can be appended without rebuilding earlier roots. Verifiable random function-based ordering proofs can establish the order of transactions without requiring block-level hashes. Threshold signatures from a validator set over recent checkpoints can serve a similar role. Each of these mechanisms tamper-evidently commits to an ordering of transactions without requiring the block-by-block hash chain that defines contemporary blockchain design.

6.5 State Management Becomes Optional

The most consequential simplification is to state management. In contemporary chains, the chain stores world state: every account balance, every contract storage slot, every UTXO. The chain literally maintains a giant Merkle Patricia trie (in Ethereum) or a UTXO set (in Bitcoin), and most of the chain's complexity and storage cost arises from this responsibility.

In Settlement Ether, the chain need not maintain state at all. State can be *derived*, by anyone who cares to compute it, from the ordered log of transactions. Three implementation strategies are available.

Stateless chain with state proofs. The chain commits only to a root hash of the implied state. Users include state proofs (Merkle proofs against the committed root) when they spend, and the chain validates the proofs against its committed root before applying the transaction's effects. State lives off-chain, maintained by users and indexers. The chain emits a new state root after each transaction; the trie itself is not stored on chain.

Receipt-based balance tracking. Each transaction produces a receipt: a signed statement of the new state of the affected accounts. Receipts are stored on chain in an indexed structure; the chain itself does not aggregate them into a global state. To find a balance, one looks up the most recent receipt for the address in question.

Pure event log with off-chain rollup. The chain is an event log; users (and indexers, and exchanges) consume the log and maintain local state. The chain commits only to a Merkle root of the implied state, periodically; the state itself is reconstructed by whoever needs it.

In all three strategies, the chain has shed its responsibility for being the *source* of state. State is derived from the ordering; the chain produces the ordering.

6.6 The Resulting System

The resulting system is recognizably descended from Bitcoin and Ethereum but architecturally distinct from either. It is layered, modular, and proof-mediated. The base layer is a settlement zone, not a transaction zone. The transaction zone is the layers above. The base layer maintains a totally-ordered log with cryptographic tamper-evidence; state is derived; smart contracts execute on rollups; ownership is asserted by sovereign keys held by users.

The system has three properties worth emphasizing.

It is *quantum-resistant by construction*, because the only quantum-vulnerable cryptographic primitive in the entire architecture is the wallet-key signature used for network validity, which the dual-authority model has rendered insufficient for unauthorized spending—absolutely so for funds at rest and against classical theft, with the owner's transient broadcast window the sole residual exposure on chains whose script cannot yet bind the ownership proof to the transaction.

It is *architecturally minimal*, because the chain has shed responsibility for ownership state, world state, and execution. The chain is closer to a distributed log than to a contemporary blockchain.

It is *operationally efficient*, because the per-transaction cost on the base layer is dramatically reduced. The chain validates a wallet-key signature (a few microseconds), appends the transaction to a log (constant cost), updates an accumulator (logarithmic cost), and emits the new commitment. The state management, the contract execution, the gas accounting, the trie updates—none of these are responsibilities of the base layer.

7 Throughput Analysis

The throughput implications of the Settlement Ether architecture are substantial and follow directly from the architectural simplifications described in Section 6.

7.1 Bottlenecks of Contemporary Chains

Current Ethereum mainnet processes approximately 15 transactions per second on the base layer [39]. The bottlenecks are well-understood: EVM execution dominates per-transaction cost (every full node executes every contract); state-trie I/O is expensive (each transaction reads and writes multiple slots of state); gas accounting must be applied to every operation; signature verification, while individually fast (approximately 30 microseconds for ECDSA verification on commodity hardware), accumulates over thousands of signatures per block; block propagation is bandwidth-limited; consensus messages (BLS attestations from tens of thousands of validators per slot) are themselves a substantial fraction of the chain's bandwidth.

7.2 Bottlenecks Removed Under Settlement Ether

Under the Settlement Ether architecture, several of these bottlenecks are eliminated entirely. EVM execution is removed (contracts execute on rollups, not on the base layer). State trie I/O is removed (state is derived, not stored). Gas accounting is simplified to a per-transaction fee schedule. The signature verification load is reduced (the chain verifies only the wallet-key signature; the ownership-key signature is interpreted off-chain by parties who need to derive balances). Block-level consensus messages are reduced (BFT consensus over a smaller validator set replaces large-set BLS aggregation, or proof-of-history-style ordering replaces block-level consensus entirely).

The per-transaction cost on the base layer, in the limit, reduces to (i) signature verification

for the wallet key, (ii) append-to-log operation, (iii) accumulator update, and (iv) emission of the new commitment. The first is a constant operation; the second is constant; the third is logarithmic in log size; the fourth is constant. The total per-transaction work is approximately $O(\log n)$ where n is the log size.

7.3 Quantitative Estimates

A rigorous benchmark of the Settlement Ether architecture awaits implementation, but rough order-of-magnitude estimates can be derived from comparison with existing systems. Solana, which uses proof-of-history-based ordering and a relatively simplified execution model, achieves approximately 3,000 to 4,000 transactions per second in production [34]. Sui, which uses an object-centric model with parallel execution, claims theoretical throughput in the range of 100,000 to 150,000 transactions per second [35]. Pure log-replicated state machines, used in production database systems, routinely achieve hundreds of thousands of transactions per second under appropriate hardware assumptions.

Settlement Ether's base layer, with state management and execution stripped to higher layers, plausibly achieves throughput in the range of tens of thousands to one hundred thousand transactions per second on commodity hardware, bounded primarily by signature verification throughput and network bandwidth. This is two to four orders of magnitude greater than contemporary Ethereum mainnet, and exceeds the throughput of any chain currently in production.

The throughput at higher layers (rollups, app-chains) is bounded by different constraints—primarily the cost of producing zero-knowledge proofs for rollup transitions—but is, in current rollup architectures, already orders of magnitude greater than the base layer. The aggregated throughput of a Settlement Ether system, with many rollups settling to the base layer, is in principle bounded only by the global capacity to produce settlement proofs.

7.4 Comparison with Alternative Throughput Approaches

Several alternative approaches to blockchain throughput exist. Rollup-centric Ethereum [16] moves execution off the base layer while retaining the base layer's role as ownership-truth source; this improves throughput but does not eliminate the state-management bottleneck. Sharding splits the chain into multiple parallel chains, each maintaining its own state; this improves aggregate throughput but introduces cross-shard coordination costs. Modular blockchains separate execution, settlement, and data availability into different layers; this is closer in spirit to Settlement Ether, but typically retains the chain's role as ownership authority.

Settlement Ether differs from all of these in that it relaxes the chain's responsibility for ownership truth entirely. The throughput improvement is, accordingly, larger and more fundamental: it comes from the chain having less to do, not from doing the same work more efficiently.

8 The Three Generations of Cryptocurrency

We can now position Settlement Ether within the broader trajectory of cryptocurrency evolution. Three generations are distinguishable, each defined by a categorical change in what the chain is for.

8.1 Generation 1.0: Bitcoin and Cryptographic Value

Bitcoin's contribution was the demonstration that value could be made cryptographic—that a distributed, censorship-resistant ledger of value could be maintained without any central author-

ity. The chain's role was definitive: the chain established who owned what. The cryptographic mechanisms (UTXOs, hash-chained blocks, proof-of-work) all served this purpose. Bitcoin's architectural commitment is that the chain is the source of ownership truth, and that the truth itself is established by cryptographic procedures rather than by institutional decree.

The Generation 1.0 paradigm produced approximately a trillion dollars of recognized economic value over its first sixteen years, and inaugurated the cryptocurrency industry as a whole.

8.2 Generation 2.0: Ethereum and Programmable Value

Ethereum's contribution was the demonstration that the same techniques could be extended from value to general computation. By placing a Turing-complete virtual machine on the chain, Ethereum enabled programmable contracts that could express arbitrary logic. From this emerged the decentralized application ecosystem: decentralized finance, non-fungible tokens, decentralized autonomous organizations, decentralized identity systems, prediction markets, on-chain games, and many others.

The architectural commitment of Generation 2.0 is, structurally, the same as Generation 1.0's: the chain is the source of ownership truth, and the truth is established by cryptographic procedures. What changes is the expressive range of those procedures: instead of fixed transaction patterns, Generation 2.0 supports arbitrary programs.

The Generation 2.0 paradigm produced the decentralized application ecosystem and the technical and conceptual infrastructure for the rollup-centric roadmap, account abstraction, and modular blockchain design.

8.3 Generation 3.0: Settlement Ether and Sovereign Value

Settlement Ether's contribution is the demonstration that ownership truth can exist independently of the chain—that the cryptographic establishment of ownership need not be supplied by the chain's native primitives, and that the chain can be reduced to a settlement and ordering service without losing the security properties that make cryptocurrency valuable.

The architectural commitment of Generation 3.0 is, structurally, a relaxation of Generation 1.0's and 2.0's: the chain is not the source of ownership truth. Truth is sovereign. It is established by the holders of ownership keys, drawn from cryptographic schemes that the chain need not understand. The chain provides ordering and settlement; ownership is a property held by users.

This is a categorical change, comparable in magnitude to the change from Generation 1.0 to Generation 2.0. The architectural minimization that follows—block structure becoming optional, state management becoming optional, hash chaining becoming optional—is the natural consequence of the chain having a smaller responsibility.

The Generation 3.0 paradigm has, as of mid-2026, been demonstrated in production through the Lemnicoin deployment and in patent specification through the Bitcoin Tapscript construction. The full ecosystem of applications that Generation 3.0 will support—quantum-resistant decentralized finance, sovereign-identity systems, ledger-agnostic ownership, recoverable value custody—is still being established.

8.4 Continuity Across Generations

We emphasize that each generation includes its predecessors as a special case. A Generation 3.0 chain can host Generation 2.0 smart contracts (on rollups or other higher layers); a Generation 2.0 chain can be used as a Generation 1.0 store of value. The generations are not mutually exclusive; they are nested.

The migration from Generation 2.0 to Generation 3.0 does not require abandoning Generation 2.0 capabilities. It requires extending the cryptographic stack to include ownership separation, which can be done without disturbing existing Generation 2.0 applications. The Lemnicoin implementation demonstrates this: LEMNI is a Generation 3.0 token (with separated ownership keys) deployed on a Generation 2.0 chain (Polygon, EVM-compatible). The token's behavior is fully compatible with existing DeFi infrastructure—decentralized exchanges, lending platforms, wallets—while providing Generation 3.0 ownership guarantees to its holders.

9 Migration Paths

The migration from Generation 2.0 to Generation 3.0 admits three principal paths, each with distinctive properties and use cases.

9.1 License-and-Overlay

The most direct path, and the one we recommend, is for an existing Generation 1.0 or 2.0 chain to license cryptographic ownership separation technology and adopt it as the standard ownership-validation layer for its tokens. The chain itself does not undergo any protocol change; the ownership separation operates as an application-layer overlay (in the EVM case) or as a transaction-construction convention (in the Bitcoin Tapscript case). Users opt into ownership separation by constructing dual-authority outputs; users who do not opt in retain their current Generation 2.0 behavior.

Under this path, the chain progressively transforms into a Generation 3.0 chain not by changing its protocol but by changing the dominant transaction patterns. As more users adopt ownership separation, more of the chain's transactional value becomes ownership-separated, and the chain's role for those users becomes that of a settlement layer rather than an ownership-truth source. Over time, the chain's architecture can be evolved (through normal governance and protocol-upgrade mechanisms) to better serve the settlement-layer role: state management can be lifted off, execution can be migrated to rollups, consensus can be simplified, throughput can be increased.

This is the path most consistent with the conservative-change ethos of major blockchain communities, particularly Bitcoin's. It permits incremental adoption, preserves backward compatibility, and does not require any consensus-level decision.

9.2 Hard Fork

An alternative path is for a chain's governance to adopt cryptographic ownership separation as the standard ownership-validation mechanism at the protocol level, through a coordinated hard fork. Under this path, the chain's native signature scheme remains in place (to validate wallet-key signatures), but the chain protocol also requires ownership-key signatures for certain categories of transactions—perhaps for high-value transactions, perhaps for all transactions above a configurable threshold, perhaps for all transactions after a specified flag day.

This path provides stronger security guarantees by making ownership separation mandatory rather than opt-in, but requires the political coordination that protocol-level changes entail. For most major chains, this path is feasible but slow; for chains with active governance mechanisms (Tezos, Cardano, Polkadot), it may be the natural choice.

9.3 Greenfield Deployment

A third path is to design new chains from the genesis block as Generation 3.0 chains, with cryptographic ownership separation built in at the protocol level and the chain optimized for the settlement-layer role from the start. Such chains can incorporate the lessons of the Generation 3.0 architectural inversion without compromise: minimal state, log-based ordering, BFT consensus over a small validator set, native quantum-resistant primitives, and a base layer optimized for settling proofs from higher layers rather than for direct transaction execution.

Greenfield chains do not solve the problem of migrating existing value held on Generation 1.0 and 2.0 chains, but they offer destinations to which value can migrate and competitive pressure that may accelerate the license-and-overlay adoption on existing chains.

10 A Worked Greenfield Deployment: End-to-End Mechanics

The preceding section described greenfield deployment in the abstract. This section makes it concrete. We trace a single payment through a hypothetical Generation 3.0 chain—call it *SE-0*—from the moment value is deposited to the moment a recipient independently confirms it, exhibiting at each step exactly what the chain stores, what it checks, and what it deliberately does not do. The reader who wishes to see why the architecture is at once more streamlined and harder to defeat than its predecessors will find the whole argument here in miniature. Every simplification below is a direct consequence of one decision: cryptographic proof of ownership has been moved off the chain.

10.1 The two keys

Each holder controls two structurally distinct keys, as in the dual-authority model of Section 4. The *wallet key* is an ordinary, possibly quantum-vulnerable key used only for network-validity purposes—rate-limiting, anti-spam, and optionally fee payment. The *ownership key* is a post-quantum signature key (Falcon, Dilithium, or SPHINCS+/SLH-DSA) and is the sole authority that can authorize the movement of value. Both derive deterministically from a single seed, so the holder manages one secret.

10.2 What lives on chain: commitments, not keys

When value is assigned to a holder, what is written to SE-0 is not a public key but a *commitment* to one: a hash $C = \mathrm{Commit}(pk_{\mathrm{own}})$ of the post-quantum ownership public key (and, where the wallet-key check is enforced on chain, of the wallet public key as well). No public key appears in the clear while the value sits idle. This single choice is the source of quantum-safety at rest: a harvester scanning the ledger for unspent value finds only hashes, and a hash exposes nothing for Shor's algorithm to attack. The ownership public key—and hence anything a quantum adversary could exploit—enters the record only in the witness of the spending transaction, and then, as shown below, only in a form bound irrevocably to that one transaction.

10.3 The ledger as a hash-linked transaction stream

SE-0 has no blocks. The ledger is an append-only stream of individual transactions, each cryptographically linked to its predecessor:

$$h_i \;=\; H\big(\,\mathrm{payload}_i \,\|\, \sigma_i \,\|\, h_{i-1} \,\|\, R_i\,\big),$$

where h_{i-1} is the hash of the previous transaction and R_i is the state-root commitment produced after transaction i is applied. The chain of hashes supplies tamper-evidence—altering any historical transaction changes its hash and breaks every subsequent link—and embedding R_i makes the stream self-auditing: verifying history is replaying each transaction and checking that the transition $R_{i-1} \to R_i$ is exactly what its inputs and outputs imply. A single Byzantine-fault-tolerant sequencer (or a small BFT validator set) assigns positions in the stream and signs them, establishing the total order; periodically a Merkle Mountain Range [38] root is published so that inclusion and state proofs remain logarithmic in length. None of this requires proof-of-work, proof-of-stake, or aggregation of transactions into blocks: the only thing agreed is order.

10.4 Anatomy of a transaction

A payment carries (i) its *inputs*, each naming a prior output together with that output's amount, its ownership commitment, and a Merkle *state proof* of its existence and unspent status against a recent finalized root; (ii) its *outputs*, each a fresh ownership commitment and an amount; (iii) a *fee*; (iv) a *freshness anchor*—a recent finalized state root R the holder read before signing; (v) the *ownership authorization* σ; and (vi) the link h_{i-1}. The recommended construction is non-interactive: the holder simply reads the latest finalized root and binds it in, with replay prevented in the UTXO model by the one-time consumption of each input (an account model would add a per-owner counter). An interactive variant—reserving the input under a single-use ticket before revealing—is available where instant pre-confirmation or atomic multi-party spends are wanted, but it is not required for security.

10.5 Creating and authorizing a payment

The holder assembles the transaction body T_{core} (all inputs, all outputs, all amounts, the fee, and the anchor R) and signs it with the ownership key:

$$\sigma \;=\; \text{Sign}_{pk_{\text{own}}}(H(T_{\text{core}})).$$

Because σ is a message-bound signature over the entire body, it authorizes *this transaction and no other*: the destination, the amounts, and the anchor are all inside the signed digest. The signature is computed entirely offline; no interaction with the chain is required to produce it.

10.6 Admission: how the chain decides a transaction is valid

When the transaction arrives, the sequencer evaluates a fixed predicate. Each check is constant- or logarithmic-time, and all must pass:

1. **Input authenticity.** Each input's state proof verifies against the committed root, establishing that the output exists, is unspent, and carries the stated amount and ownership commitment. The amount cannot be fabricated: the proof must match the root the whole network already agrees on.
2. **Ownership authorization.** The revealed ownership public key satisfies $\text{Commit}(pk_{\text{own}}) = C$ for the input's commitment, and $\text{Verify}(pk_{\text{own}}, H(T_{\text{core}}), \sigma)$ holds.
3. **Conservation.** $\sum$ input amounts $= \sum$ output amounts + fee. Value cannot be created.
4. **No double-spend.** Each input is currently unspent; it is marked consumed atomically on append.
5. **Freshness and anti-spam.** The anchor R is recent and finalized, and the inexpensive wallet-key check passes.

If the predicate holds, the transaction is appended, its inputs are marked spent, its outputs are inserted, and a new root R' is emitted. If any check fails, the transaction is never admitted—it does not enter the permanent record at all. Notice what is *absent*: no smart-contract execution, no gas metering, no world-state trie traversal, no block assembly, no proof-of-work or proof-of-stake. The chain verifies self-contained mathematics and records an order.

10.7 Why forgery, theft, inflation, and replay all fail

The construction is bullet-proof against the obvious attacks not by vigilance but by structure:

- *Forgery* requires the post-quantum ownership secret; the on-chain commitment reveals nothing usable, and the signature scheme is quantum-resistant.
- *Tampering* with the amount or destination invalidates σ, which is bound to the entire body.
- *Inflation* is rejected by the conservation check, and *spending a non-existent input* fails the state proof.
- *Replay* is impossible because the input has already been consumed.
- *Front-running during broadcast*—the one residual exposure of the retrofit Bitcoin construction (Section 4)—is closed here natively. Because σ is message-bound, the ownership material revealed at spend authorizes only this transaction; an adversary who harvests it cannot redirect funds. Recovering the wallet key buys nothing, since under dual authority the wallet key alone cannot move value.

10.8 Verifying an incoming payment

A recipient need not run a full node or trust one. Everything required to validate an inbound payment travels with it or is a short proof: the recipient checks σ against the revealed ownership key and its commitment, checks the inputs' state proofs against a finalized root, and checks the sequencer's signatures placing the transaction at a final position in the order. The work is logarithmic in the size of the ledger and runs on a phone. Crucially, the recipient trusts the validator set only for *ordering and finality*—a small, checkable set of signatures—and trusts no one for *ownership*, which they verify themselves. This is a strictly smaller trust assumption, and a strictly smaller computation, than safely accepting a payment on a contemporary chain.

10.9 What other chains carry that SE-0 does not

Table 1 sets the example chain beside Bitcoin and Ethereum. The pattern is uniform: every component SE-0 omits is one whose purpose was to make the chain the authority on ownership. Once that authority is moved off-chain, the component has nothing left to do.

10.10 The throughput consequence

The base-layer cost of a transaction on SE-0 is: verify a signature, check a logarithmic state proof, append one entry, update an accumulator, and emit a root. There is no global state to serialize against, so admission is parallelizable; there is no execution and no trie I/O, which are the costs that actually bound contemporary chains; and consensus carries only an ordering decision. The result is the two-to-four-orders-of-magnitude headroom argued in Section 7—not won by cleverness, but by the chain having almost nothing left to do. The throughput, the minimality, and the resistance to forgery are not three achievements; they are one achievement

Aspect	**Bitcoin (Gen 1.0)**	**Ethereum (Gen 2.0)**	**Greenfield SE (Gen 3.0)**
Unit of consensus	Block	Block	None: hash-linked transaction stream
Consensus mechanism	Proof-of-work	Proof-of-stake	BFT ordering over a small validator set
What consensus decides	Block validity + order	Validity + execution + order	Order, availability, finality only
Global world state on chain	Full UTXO set	Full Merkle–Patricia trie	None: only a state-root commitment
Per-node state growth	UTXO-set growth	Unbounded full state	$O(1)$ at base layer; state derived off-chain
Execution at base layer	None (limited script)	EVM, gas-metered	None: execution on higher layers
Authority over ownership	Chain's ECDSA	Chain's ECDSA	Off-chain, user-held PQ ownership key
Public key on chain at rest	Hash, but key reused/exposed in practice	Revealed on first spend	Never: only a hash commitment
Quantum-vulnerable key able to move funds	ECDSA (sufficient alone)	ECDSA (sufficient alone)	None: wallet key insufficient; ownership key is PQ
Accepting an incoming payment	Full node or trust	Full node or trust	Recipient, locally, from an $O(\log n)$ proof
Double-spend resolution	Longest-chain confirmations	Finality gadget	Total order + input consumption at admission
Approx. base-layer throughput	∼7 tps	∼15–100 tps	Orders of magnitude higher (parallelizable)

Table 1: What the first two generations carry that a greenfield Settlement Ether chain does not. Each omitted component existed to make the chain the source of ownership truth; moving ownership off-chain removes its reason to exist.

seen from three sides, and the achievement is the relocation of cryptographic proof of ownership off the chain.

10.11 Confidentiality and the privacy default

The construction described so far is *transparent*: amounts appear in the clear—the conservation check requires them—and the input–output graph is visible, so the ledger is pseudonymous in exactly the sense Bitcoin is, and is subject to the same deanonymization by graph analysis. Cryptographic ownership separation is orthogonal to confidentiality: it relocates the *authority* that moves value, not the *visibility* of the value moved. A Settlement Ether chain therefore inherits the discretion problem of its predecessors unless confidentiality is designed in. What this subsection establishes is that the settlement-layer architecture is unusually well-positioned to design it in—more so than an execution-centric Generation 2.0 chain.

The reason is structural. The base layer does nothing but order transactions and verify proofs; it never executes contracts and never needs to *read* the values it settles. Every transparent field can therefore be replaced by a hiding commitment accompanied by a zero-knowledge proof, and the shape of the admission predicate is unchanged. The transparent checks of the preceding subsections map almost one-to-one onto the shielded transaction model pioneered by Zcash, as Table 2 shows.

Transparent admission check (above)	**Zero-knowledge (shielded) realization**
Input exists and is unspent (Merkle proof against the state root)	Zero-knowledge proof of membership in a note-commitment tree, revealing which note to no one
Ownership authorization (Verify(pk_{own}, σ))	Spend authority proven inside the zero-knowledge statement
Conservation ($\sum$ inputs = $\sum$ outputs + fee)	Homomorphic balance over hiding commitments, with range proofs
No double-spend (mark input consumed)	Nullifiers: a deterministic tag marks a note spent without revealing which

Table 2: The transparent admission predicate developed above maps one-to-one onto a shielded, zero-knowledge construction, precisely because the settlement layer verifies validity without needing to read transaction contents.

In an execution-centric architecture, confidentiality is a bolted-on overlay that must contend with a virtual machine requiring plaintext state in order to compute. In a settlement-only architecture the base layer already declines to inspect contents, so "shielded by default, with the validator verifying validity in zero knowledge" is the natural design rather than a retrofit.

Discretion in the literal sense follows from the same construction through *viewing keys*. A recipient scans the stream for notes addressed to them; an auditor, counterparty, or regulator learns of a holder's activity only when the holder surrenders a viewing key. Confidentiality from the public is thus combined with selective, holder-controlled disclosure—an inversion of the present default, under which all activity is visible to everyone, into one under which nothing is visible unless the owner permits it.

Two qualifications must be stated plainly. **First**, confidentiality and post-quantum security must be achieved *together*, and the efficient privacy primitives in production use today—Pedersen commitments, Groth16, Bulletproofs—are built on elliptic curves and are not quantum-resistant. Layering them onto a Settlement Ether chain would reopen, on the confidentiality axis, precisely the quantum exposure that ownership separation closes on the authorization axis. Keeping both properties requires post-quantum-friendly machinery: hash-based STARKs, which are transparent (requiring no trusted setup) and post-quantum by assumption, for the

proofs, and lattice-based commitments (resting on SIS/LWE hardness) for hidden amounts that still admit a homomorphic conservation check. Such a design is coherent and avoids a trusted setup, but its proofs are larger and more costly than curve-based systems, and post-quantum confidential-amount commitments remain an area of active research. Confidentiality is therefore available to a Settlement Ether chain, but quantum-safe confidentiality carries a real and, at present, heavier cost.

Second, the ledger's cryptography does not by itself close *metadata* leakage. A shielded transaction hides amounts and links even from the sequencer—which verifies validity in zero knowledge without learning contents—but the existence, timing, and network origin of a submission remain observable, and the ordering itself is public. Closing this channel requires mechanisms orthogonal to the ledger: encrypted or threshold mempools that withhold transaction contents until ordering is fixed, and network-layer mixing. These are complementary to, not substitutes for, the on-chain construction.

In sum, a greenfield Settlement Ether chain does not enforce privacy, and in its transparent form it carries the same discretion problem as existing chains. But because the settlement layer only orders and verifies proofs and never reads state, it removes the structural obstacle that an execution-centric chain faces, making shielded-by-default operation with holder-controlled disclosure a natural rather than adversarial design choice—at the price of adopting post-quantum-friendly zero-knowledge primitives, so that confidentiality is not purchased by surrendering the quantum resistance that motivates the architecture in the first place.

11 Discussion and Related Work

11.1 Comparison with Quantum-Safe Transaction Schemes

The most direct points of comparison for cryptographic ownership separation in the Bitcoin context are the Quantum-Safe Bitcoin (QSB) proposal by Levy and the StarkWare team [26], and Linus's Binohash construction [27]. Both QSB and Binohash address the related but distinct problem of making Bitcoin transactions unforgeable by a quantum adversary. They protect transaction integrity—preventing an attacker from modifying a broadcast transaction or forging a new transaction with the same UTXO—using Lamport signatures over transaction digests, verified within Bitcoin Script.

Neither QSB nor Binohash separates ownership from the wallet key. In both schemes, the holder of the wallet private key retains full authority to create and spend transactions. If the wallet private key is stolen, the attacker can construct their own QSB- or Binohash-protected transaction to spend the funds. The schemes protect the transaction from modification, not the owner from theft.

Cryptographic ownership separation solves a different and complementary problem: ensuring that possession of the wallet key alone is insufficient to spend funds. The two constructions can be combined: a dual-authority output can itself use QSB- or Binohash-style protection for the wallet-key portion, providing both theft protection (via the ownership key) and transaction integrity (via the wrapping scheme). More generally, the transient exposure of the wallet-key portion during the owner's own broadcast is not unique to the dual-authority construction but is shared by every Bitcoin spend; beyond combination with QSB it can be mitigated operationally by submitting the spend out-of-band to an identifiable miner rather than to the public mempool—reducing the set of parties able to observe and act on the revealed witness to a single accountable operator—and is eliminated natively once `OP_CHECKSIGFROMSTACK` or `OP_CAT` permits the ownership proof itself to be bound to the spending transaction (Section 4).

11.2 Comparison with Account Abstraction

Ethereum's account abstraction proposals (ERC-4337, EIP-7701) provide a different form of authorization flexibility: a smart contract wallet can implement arbitrary verification logic, including multi-signature schemes, time-locks, biometric attestations, and recovery mechanisms. The cryptographic ownership separation model is, in one sense, a special case of account abstraction: the dual-authority requirement could be implemented as a smart contract wallet's verification policy.

The differences are operational rather than theoretical. Account abstraction operates at the smart-contract-wallet layer and requires the user to use a smart-contract wallet (incurring the deployment and per-operation costs that entails). Cryptographic ownership separation operates at the token-contract layer, with the dual-authority check enforced by the token contract itself, which means it applies to any wallet that holds the token regardless of whether the wallet is a smart contract or an externally-owned account. This makes the construction lighter-weight and more universally applicable.

Cryptographic ownership separation also extends naturally to UTXO-based chains where account abstraction has no analogue. The Tapscript-based Bitcoin construction has no equivalent in the account-abstraction literature, because account abstraction is fundamentally an account-model concept.

11.3 Comparison with Centralized Recovery and Custody

A separate family of approaches addresses the wallet-theft problem through centralized recovery mechanisms: custodial wallets, multi-signature services, MPC-based key management. These approaches provide theft protection by introducing trusted intermediaries who can override transactions or recover lost keys.

Cryptographic ownership separation provides the same theft protection without introducing intermediaries. The ownership key is held by the user; no third party can override a transaction; no third party can recover the ownership key for the user. The construction is fully sovereign: the user retains full self-custody while gaining protection against wallet-key theft.

11.4 The Three Architectures in Concert

We have argued that Settlement Ether represents a third evolutionary stage of cryptocurrency. This does not imply that the three generations are mutually exclusive or that Generation 3.0 supplants its predecessors. The most likely future is one in which all three architectures coexist: Generation 1.0 chains continue to serve as stores of value; Generation 2.0 chains continue to support the decentralized application ecosystem; Generation 3.0 chains serve as the settlement-layer substrate for both, providing quantum-resistant ownership guarantees and high-throughput infrastructure.

The migration is not a replacement but an extension. A Generation 3.0 chain can host Generation 2.0 contracts on rollups; a Generation 2.0 application can use Generation 3.0 ownership for its tokens. The categorical distinctions are useful for analytical purposes; in production, the architectures will blend.

12 Conclusion

This paper has argued that the post-quantum migration of blockchains, conventionally framed as a protocol-coordination problem at the chain level, is more naturally understood as a

cryptographic-construction problem at the user level. The quantum threat to a blockchain reduces, in practice, to a single primitive: the signature scheme that authorizes value transfer. By separating that signature into two structurally distinct authorities—a wallet key for network validity and an ownership key for actual authorization—the quantum-resistance problem can be solved unilaterally by individual users, on any chain that supports programmable spending conditions, without any change to the chain's protocol.

The construction has been demonstrated in production by the Lemnicoin deployment on Polygon mainnet, and has been extended to Bitcoin via Tapscript in a continuation-in-part patent filing (European Patent n°EP26300368.3, 2026). The underlying technique—cryptographic ownership separation—is independent of the specific post-quantum signature scheme adopted; Falcon, Dilithium, SPHINCS+, and other quantum-resistant primitives all serve.

The architectural implications go beyond quantum resistance. Once the chain has been relieved of the responsibility for validating ownership, the chain can be radically simplified. State management becomes optional. Block structure becomes optional. Hash chaining becomes optional. The chain becomes a settlement layer in the precise sense: a totally-ordered, tamper-evident log of transactions whose ownership is established off-chain. We have called this resulting architecture *Settlement Ether*, and have argued that it constitutes the third evolutionary generation of cryptocurrency. Bitcoin (Generation 1.0) demonstrated cryptographic value. Ethereum (Generation 2.0) demonstrated programmable value. Settlement Ether (Generation 3.0) demonstrates sovereign value—value whose ownership is established by the holder, not by the chain.

The throughput consequences are substantial: estimated improvements of two to four orders of magnitude over contemporary base-layer chains, achievable not by clever optimization of existing work but by the chain having less work to do in the first place.

The migration to Generation 3.0 does not require a coordinated event. It proceeds, transaction by transaction, as individual users adopt dual-authority outputs. The chains that license and adopt cryptographic ownership separation will undergo a gradual but categorical architectural transformation, the end-state of which is a Settlement Ether system whose relationship to contemporary blockchains is one of genealogical, not architectural, continuity.

The clock has been running on the quantum migration for the entire history of every chain in current use. The post-quantum future of blockchain is not, as conventionally assumed, a future of coordinated protocol changes and reluctant validator upgrades. It is a future of sovereign ownership keys, of chains that have shed their responsibility for ownership truth, and of settlement layers that do less so that they can do it much faster. This is the architectural inversion that cryptographic ownership separation makes possible, and that Settlement Ether is the natural expression of.

Author's Closing Remarks

Every architecture encodes a relationship of power, and it is worth saying plainly what the prevailing blockchain architecture quietly made of that relationship. When the chain is the source of ownership truth—when what you hold is held only because the chain, consulted, attests that you hold it—then your ownership is not yours. It is granted. It is a tenancy, and a freedom that must be perpetually re-granted by an authority is not freedom but permission.

It would be easy to overclaim what has changed here, so let me be exact about what has not. We have not abolished the private key, nor the act of proving possession of it. A signature is still a signature; the holder still guards a secret and still demonstrates control of it. In that narrow mechanical sense nothing is new—proving ownership with a key off the chain looks just like proving it on the chain, and a quantum-resistant key proves possession no differently than the key it replaces. The novelty is not in the instrument. It is in the relationship the instrument

now stands in.

Two independent things are at work, and they should not be confused. The move to a post-quantum ownership key is a stronger lock—it guards the secret against an adversary the old lock could not withstand. But a stronger lock on a tenant's door does not make the tenant an owner. The change that confers ownership is the other one: where the deed is kept, and who is permitted to rule on it. The lock is cryptography; sovereignty is jurisdiction.

In the first two generations, the chain's own signature scheme, under the chain's own consensus rules, was the very definition of ownership. The chain was not merely where your title was recorded; it was the authority that decided what counted as title at all—and an authority that defines ownership can redefine it: by fork, by migration, by freeze, by blacklist, by the quiet pressure brought to bear on whoever controls the rules. Your key was a pass the chain agreed to honor, contingent on rules you did not write. Cryptographic ownership separation moves the deed off the chain entirely. Ownership is constituted by a secret under cryptography the holder chooses, committed off-chain; the chain holds no rule that defines it and therefore no lever that can alter it. The key stops being a pass and becomes the deed itself.

It is fair to ask whether the chain truly has no say. Over ownership, it does not: it cannot decide that you do not own what your secret controls, cannot reassign it, cannot fabricate a rival claim, cannot freeze it. Where it checks anything at all, it verifies a proof you supply against a commitment you set—a notary confirming your mathematics, never a judge with discretion to rule against it. One power does remain, and honesty requires naming it: the chain still decides whether, and in what order, to record a transaction. It can refuse to witness your move—it can censor. But censorship is a power over action, not over ownership; it can delay your spending, never undo your owning. And because the deed lives off the chain and is bound to no single ledger, even that power is escapable: ownership is not the chain's hostage, and what one recorder refuses to witness, another will. The chain may decline to act; it can no longer presume to own.

This is what *sovereign* value means, and the word is not decoration. When ownership is constituted by a secret only its holder possesses, no chain, validator, custodian, or intermediary holds the authority to move, freeze, or reassign it—and so there is no chokepoint for a court or a government to compel. Seizure presupposes someone who can be ordered to surrender the asset; this construction leaves no such someone. The value answers to one key alone; it cannot be attached by decree, confiscated by writ, or frozen by edict, because no party exists with the power to do so—not by hiding it, but by the plain mathematics of who can act upon it. The key did not go away. What went away was the landlord's discretion over it. The blockchain of the future does not grant ownership; it merely remembers. Let those who come after hold what is theirs by their own hand, owe no fealty to the chain that records it, and mark this as the moment when cryptographic value ceased to be lent to its holders and began, at last, to belong to them.

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Patent and Intellectual Property Notice

The cryptographic ownership separation technique, the dual-authority output construction, the NUMS-internal-key construction, and the Bitcoin Tapscript implementation described in this paper are the subjects of European Patent n°EP26300368.3 (2026) [42] and the prior application [41]. Implementations of these techniques may require licensing.

References

[1] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.

[2] V. Buterin, "Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform," 2014.

[3] G. Wood, "Ethereum: A Secure Decentralised Generalised Transaction Ledger," *Ethereum Yellow Paper*, 2014.

[4] P. W. Shor, "Algorithms for Quantum Computation: Discrete Logarithms and Factoring," *Proceedings of the 35th Annual Symposium on Foundations of Computer Science*, 1994, pp. 124–134.

[5] L. K. Grover, "A Fast Quantum Mechanical Algorithm for Database Search," *Proceedings of the 28th Annual ACM Symposium on Theory of Computing*, 1996, pp. 212–219.

[6] W. Diffie and M. E. Hellman, "New Directions in Cryptography," *IEEE Transactions on Information Theory*, vol. 22, no. 6, 1976, pp. 644–654.

[7] R. L. Rivest, A. Shamir, and L. Adleman, "A Method for Obtaining Digital Signatures and Public-Key Cryptosystems," *Communications of the ACM*, vol. 21, no. 2, 1978, pp. 120–126.

[8] D. Johnson, A. Menezes, and S. Vanstone, "The Elliptic Curve Digital Signature Algorithm (ECDSA)," *International Journal of Information Security*, vol. 1, no. 1, 2001, pp. 36–63.

[9] P. Wuille, J. Nick, and A. Towns, "BIP-341: Taproot: SegWit version 1 spending rules," Bitcoin Improvement Proposal, January 2020.

[10] P. Wuille, J. Nick, and A. Towns, "BIP-342: Validation of Taproot Scripts," Bitcoin Improvement Proposal, January 2020.

[11] H. Beast, M. Heilman, and J. Foxen-Duke, "BIP-360: Pay-to-Quantum-Resistant-Hash (P2QRH)," Bitcoin Improvement Proposal, February 2026.

[12] J. Lopp et al., "BIP-361: Sunset of Legacy ECDSA Outputs," Bitcoin Improvement Proposal, April 2026.

[13] V. Buterin et al., "ERC-4337: Account Abstraction Using Alt Mempool," Ethereum Request for Comments, 2021.

[14] "EIP-7701: Native Account Abstraction with EOF," Ethereum Improvement Proposal, 2024.

[15] N. Mudge, "EIP-2535: Diamonds, Multi-Facet Proxy," Ethereum Improvement Proposal, 2020.

[16] V. Buterin, "A rollup-centric ethereum roadmap," *Ethereum Magicians forum*, October 2020.

[17] National Institute of Standards and Technology, "Post-Quantum Cryptography Standardization," announcement of FIPS 203/204/205, August 2024.

[18] NIST, "Module-Lattice-Based Key-Encapsulation Mechanism Standard," FIPS 203, August 2024.

[19] NIST, "Module-Lattice-Based Digital Signature Standard," FIPS 204, August 2024.

[20] NIST, "Stateless Hash-Based Digital Signature Standard," FIPS 205, August 2024.

[21] T. Prest et al., "FALCON: Fast-Fourier Lattice-based Compact Signatures over NTRU," submission to NIST PQC Standardization, currently designated FN-DSA (FIPS 206, in development).

[22] C. Aguilar Melchor et al., "HQC: Hamming Quasi-Cyclic," submission to NIST PQC Standardization, selected as backup KEM, March 2025.

[23] W. Castryck and T. Decru, "An efficient key recovery attack on SIDH (preliminary version)," Cryptology ePrint Archive, July 2022.

[24] W. Beullens, "Breaking Rainbow Takes a Weekend on a Laptop," Cryptology ePrint Archive, February 2022.

[25] J. Mascelli and M. Rodden, "Harvest Now, Decrypt Later: Examining Post-Quantum Cryptography and the Data Privacy Risks for Distributed Ledger Network," Federal Reserve Board, September 2025.

[26] A. Levy et al., "Quantum-Safe Bitcoin," StarkWare, April 2026.

[27] R. Linus, "Binohash: Hash-Based Bitcoin Quantum Resistance," 2026.

[28] Google Quantum AI, "Reduced Resource Estimates for Elliptic Curve Cryptanalysis," March 2026.

[29] IBM Quantum, "IBM Quantum System Roadmap," 2024–2030 projections.

[30] Quantinuum, "Helios System Announcement," November 2025.

[31] IonQ, "Roadmap to CRQC by 2028," company technical roadmap, 2025.

[32] Solana Foundation, "A Roadmap to Post-Quantum Solana," April 27, 2026.

[33] A. Yakovenko, "Solana: A new architecture for a high performance blockchain," Solana whitepaper, 2018.

[34] Solana Foundation, "Solana Network Performance Specifications," 2024.

[35] Mysten Labs, "The Sui Smart Contracts Platform," Sui Technical Paper, 2023.

[36] L. Baird, "The Swirlds Hashgraph Consensus Algorithm: Fair, Fast, Byzantine Fault Tolerance," Swirlds Tech Report, 2016.

[37] S. Popov, "The Tangle," IOTA Foundation, 2018.

[38] P. Todd, "Merkle Mountain Ranges," 2012.

[39] Etherscan, "Ethereum Network Statistics," May 2026.

[40] R. Vince, "LAIP: A Lemniscate-AGM Isogeny Protocol for Post-Quantum Ownership Authentication," Exsuperatus Technical Whitepaper (forthcoming).

[41] R. Vince, "Secure Token Ownership via Overridden Smart Contract Interface Functions with Separate On-Chain Key Binding," United States Patent and Trademark Office, USPTO Customer No. 206011, March 2026.

[42] R. Vince, "Dual-Authority Spending Conditions on UTXO-Based Blockchains via Tapscript with Separate Ownership Key Binding and Quantum-Resistant Upgrade Path," European Patent n°EP26300368.3, April 2026 (continuation-in-part of [41]).

[43] J. Mascelli and M. Rodden, "Quantum Computing Threats to U.S. Financial Infrastructure," Federal Reserve Board working paper, 2025.

[44] Citi Institute, "The Trillion-Dollar Security Race Is On: Quantum Computing and the Financial System," Citi research report, January 2026.

[45] JPMorgan Chase, "Post-Quantum Cryptography: Implications for Financial Markets," JPMorgan research note, 2025.

[46] European Commission, "Recommendation on Post-Quantum Cryptography Migration," EU Council Recommendation, 2026.